ST. JOSEPH'S
COLLEGE OF ENGINEERING AND TECHNOLOGY
- PALAI -
AUTONOMOUS

AI-Assisted Digital Forensic Investigation Platform

Project Phase I

Department of Computer Science and Engineering (Cyber Security)

Project Guide: **LIBINA ROSE SEBASTIAN**

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Introduction

- Digital forensic investigations involve collecting and analysing large volumes of heterogeneous digital evidence.
- Investigators often depend on multiple tools for artifact extraction, timeline generation, event correlation and reporting.
- Fragmented workflows make it difficult to establish relationships between events and determine the complete sequence of an incident.
- The proposed platform integrates forensic extraction, timeline reconstruction, entity relationships and intelligent analysis.
- AI is positioned as an investigation-assistance layer, not a replacement for forensic examination.

Problem Statement

Core Problem

Traditional digital forensic investigations generate large numbers of heterogeneous events and artifacts that must be manually correlated across multiple tools.

Current Challenges

- Large volumes of low-level events
- Manual timeline analysis
- Fragmented forensic tools
- Hidden artifact relationships

Investigation Requirements

- Unified evidence analysis
- Event correlation
- Incident reconstruction
- Evidence traceability

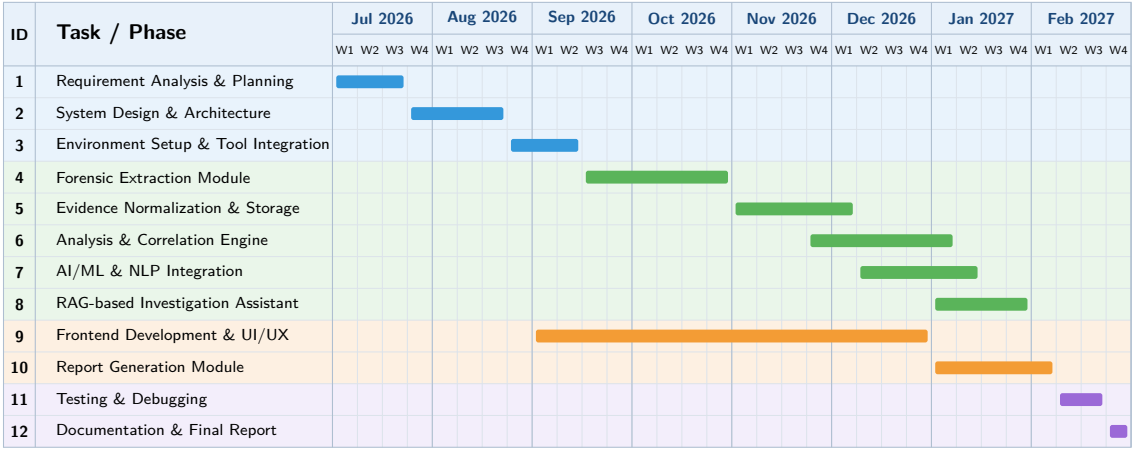
Project Gantt Chart – Phase I

Forensight AI – Digital Forensic Investigation Platform

ID	Task / Phase	Jul 2026				Aug 2026				Sep 2026				Oct 2026			
		W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4
1	Requirement Analysis & Planning																
2	System Design & Architecture																
3	Environment Setup & Tool Integration																
4	Forensic Extraction Module																
5	Evidence Normalization & Storage																
6	Analysis & Correlation Engine																
7	AI/ML & NLP Integration																
8	RAG-based Investigation Assistant																
9	Frontend Development & UI/UX																
10	Report Generation Module																
11	Testing & Debugging																
12	Documentation & Final Report																

Project Timeline

Forensight AI – Digital Forensic Investigation Platform



Literature Review – Papers 1–7

#	Paper and Source	Key Contribution	Limitation	Research Gap
1	**ForensicLLM: A Local Large Language Model for Digital Forensics** *B. Sharma et al., FSI: Digital Investigation, Vol. 52, 301872, 2025*	Locally deployable LLM for digital forensic analysis with privacy preservation and source attribution.	Covers LLM-based assistance only, not the complete forensic pipeline.	Integrate local LLM assistance with verified evidence, timelines, entity graphs and attack paths.
2	**A Filtering Model for Evidence Gathering in an SDN-Oriented Digital Forensic and Incident Response Context** *M. B. Jiménez et al., IEEE Access, 2024*	Evidence-filtering approach that reduces the volume of information investigators must examine.	Oriented to SDN and incident-response settings; no general workflow for heterogeneous evidence.	Combine filtering with disk-image analysis, artifact extraction, timelines and graph correlation.
3	**Was the Clock Correct? Exploring Timestamp Interpretation Through Time Anchors for Digital Forensic Event Reconstruction** *C. Vanini et al., FSI: Digital Investigation, 2024*	Time anchors for judging whether system timestamps represent real-world events.	Concentrates on timestamp interpretation rather than cross-artifact investigation.	Integrate timestamp validation with timeline generation, correlation and attack-path reconstruction.
4	**Creating Cybersecurity Knowledge Graphs From Malware After Action Reports** *A. Piplai et al., IEEE Access, 2020*	Knowledge graphs representing entities and relationships drawn from malware reports.	Works on reports rather than heterogeneous artifacts and chronological events.	Combine knowledge graphs with forensic artifacts, timelines and attack-path reconstruction.
5	**A Formal Model for Event Reconstruction in Digital Forensic Investigation** *S. Soltani and S. A. H. Seno, Digital Investigation, 2019*	Formal model for event reconstruction, stressing verification and reproducibility.	Becomes complex at real-world scale; no automated extraction or visualization.	Pair event reconstruction with automated extraction, visualization and AI-assisted analysis.
6	**A Complete Formalized Knowledge Representation Model for Advanced Digital Forensics Timeline Analysis** *Y. Chabot et al., Digital Investigation, 2014*	Formalized knowledge representation supporting semantic timeline analysis.	Strong formal foundation but no practical automated workflow or reporting.	Extend formal timelines with automated extraction, graph correlation and AI assistance.
7	**Emulator Forensics Investigation Model (EFIM)** *S. Şen and H. Artuner, IEEE Access, Vol. 13, 2025*	Structured, modular investigation model for emulator-based environments.	Specialized to emulator forensics; not a general-purpose platform.	Generalized modular platform for images, artifacts, timelines, graphs and AI assistance.

Literature Review – Papers 8–14

#	Paper and Source	Key Contribution	Limitation	Research Gap
8	**A Comprehensive Analysis of the Role of Artificial Intelligence and Machine Learning in Modern Digital Forensics and Incident Response** *D. Dunsin et al., FSI: Digital Investigation, Vol. 48, 301675, 2024*	Reviews AI/ML roles in forensics and incident response and the scope for automation.	Broad analysis rather than one implemented end-to-end platform.	Integrate AI with evidence management, timeline reconstruction, graphs and reporting.
9	**Artificial Intelligence in Mobile Forensics: A Survey of Current Status, a Use Case Analysis and AI Alignment Objectives** *A. Vasilaras et al., FSI: Digital Investigation, Vol. 49, 301737, 2024*	Surveys AI in mobile forensics with use cases and alignment requirements.	Mobile evidence only; no unified platform across evidence sources.	Extend AI assistance beyond mobile to multiple artifact sources and entity relationships.
10	**SIFT: Sifting File Types – Application of Explainable Artificial Intelligence in Cyber Forensics** *S. Alam and A. K. Demir, Cybersecurity, Vol. 7, 52, 2024*	Explainable AI for automated, interpretable cyber-forensic file analysis.	Scope limited to file analysis; no incident reconstruction or reporting.	Extend explainable AI to correlation, timelines, entity graphs and attack paths.
11	**Identifying Persons of Interest in Digital Forensics Using NLP-Based AI** *J. Adkins, A. Al Bataineh and M. Khalaf, Future Internet, Vol. 16, 426, 2024*	NLP techniques that surface persons of interest and investigative patterns.	Text-centric; does not by itself give cross-artifact correlation.	Combine NLP with structured artifacts, timestamps and graph-based analysis.
12	**Exploring the Potential of Large Language Models for Improving Digital Forensic Investigation Efficiency** *A. Wickramasekara, F. Breitinger and M. Scanlon, FSI: DI, Vol. 52, 301859, 2025*	Maps LLM capabilities onto forensic activities and efficiency gains.	Probabilistic output; reliability, explainability and legal concerns remain.	A controlled assistance layer working on verified structured evidence.
13	**Practitioner-Driven Framework for AI Adoption in Digital Forensics** *M. Veksler, K. Akkaya and S. Uluagac, FSI: DI, Vol. 55, 302030, 2025*	Practitioner framework for responsible, trustworthy adoption of AI.	States requirements rather than implementing a working platform.	Implement traceability, human oversight, explainable output and secure reporting.
14	**Large Language Models in Digital Forensics: Capabilities, Challenges and Future Directions** *M. Chernyshev et al., FSI: Digital Investigation, Vol. 56, 302043, 2026*	Reviews LLM capabilities, challenges and future research directions.	Probabilistic models versus deterministic, admissible forensic results.	Controlled architecture preserving forensic integrity and human oversight.

Literature Review – Overall Findings

Key Findings

- Event abstraction can reduce manual log analysis.
- Timestamp validation improves timeline reliability.
- Knowledge graphs represent complex entity relationships.
- Formal models support event reconstruction.
- AI/ML can automate forensic analysis.
- NLP supports analysis of large forensic information.
- LLMs can assist investigators and report generation.
- Explainability is important for AI-assisted forensics.

Common Limitations

- Most approaches focus on individual forensic stages.
- Limited cross-artifact correlation.
- Separate tools are often required for different tasks.
- AI approaches require validation and human oversight.
- Limited integration of timeline and graph analysis.
- Attack-path reconstruction is not consistently integrated.
- End-to-end forensic reporting is often outside the scope.

Research Gap and Proposed Contribution

Existing Approaches

- Individual forensic stages
- Large volumes of raw events
- Limited cross-artifact correlation
- Separate forensic tools
- Manual interpretation
- Limited integration of AI assistance

Our Proposed Approach

- Unified forensic workflow
- Automated artifact extraction
- Evidence integrity verification
- Timeline reconstruction
- Entity relationship graph
- Attack-path reconstruction
- AI-assisted investigation
- Secure report generation

Proposed Contribution

The proposed platform transforms raw forensic evidence into a correlated timeline, entity graph, probable attack path and explainable investigation report while using AI as an **investigation-assistance layer**.

Objectives

1. Develop a unified digital forensic investigation platform.
2. Extract artifacts from forensic disk images and system evidence.
3. Generate and normalize chronological forensic timelines.
4. Correlate events and entities using graph-based representation.
5. Reconstruct probable attack paths from correlated evidence.
6. Provide AI-assisted investigation and evidence summarization.
7. Generate structured professional forensic reports.
8. Preserve evidence integrity using cryptographic hashing and verification.

Scope of the Project

Included

- Disk image analysis
- Artifact extraction
- Timeline reconstruction
- Event correlation
- Entity relationship graphs
- Attack path reconstruction
- AI investigation assistant
- Secure report generation
- Local LLM deployment

Future / Optional

- Memory forensics
- Network forensics
- Threat intelligence enrichment
- MITRE ATT&CK mapping
- Advanced risk scoring
- Multi-investigator collaboration

Proposed System

Proposed Idea

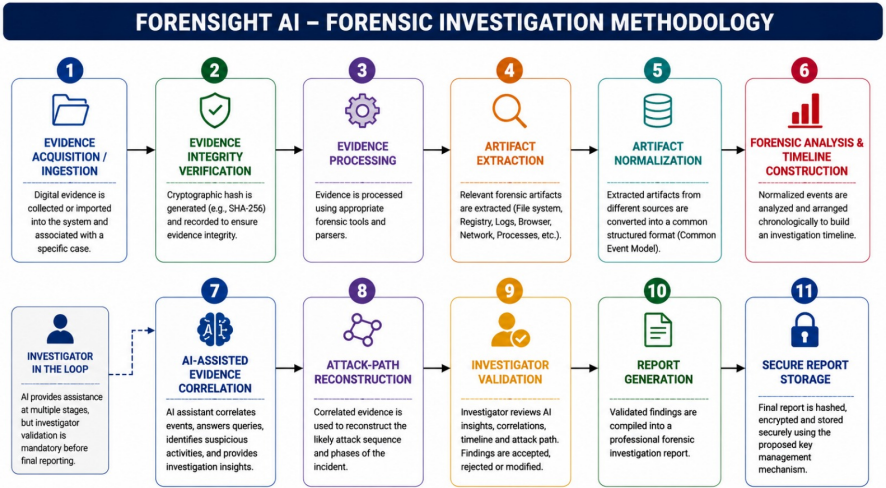
A unified and intelligent digital forensic investigation platform that transforms raw forensic evidence into a correlated timeline, entity graph, attack path and investigation report.

Digital Evidence → Artifact Extraction → Timeline Reconstruction → Entity Graph

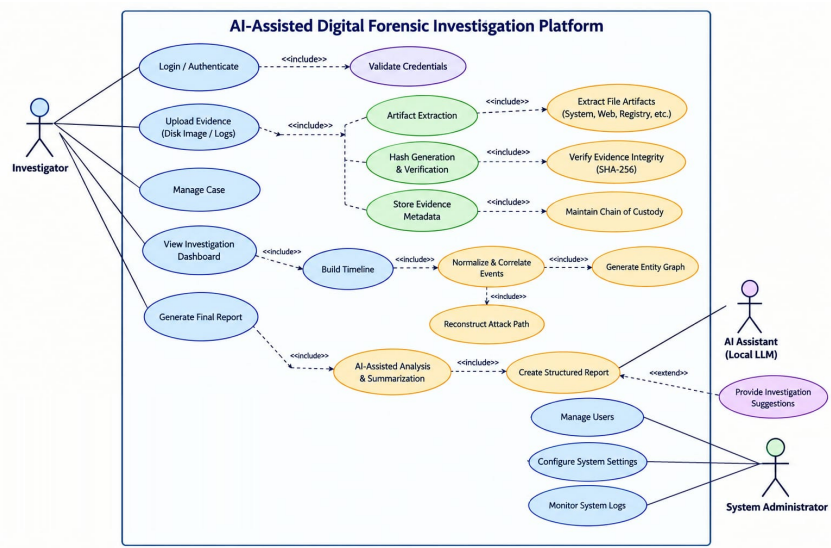
Entity Graph → Intelligent Analysis → Forensic Report

Raw Evidence → Correlated Investigation → Explainable Findings

Proposed System Methodology



UML Use Case Diagram



Major System Modules

Case & Evidence Management
Case handling and evidence management

Forensic Artifact Extraction
Extraction of relevant forensic artifacts

Forensic Analysis & Correlation
Analysis and correlation of forensic evidence

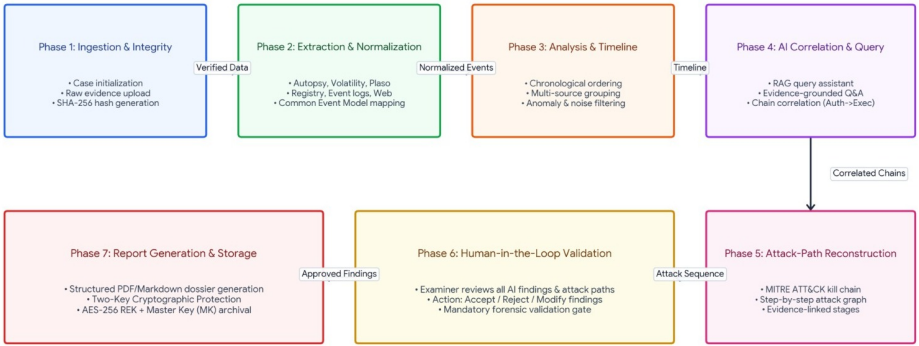
AI Investigation Assistant
AI-assisted investigation and evidence analysis

Attack-Path Reconstruction
Reconstruction of probable attack paths

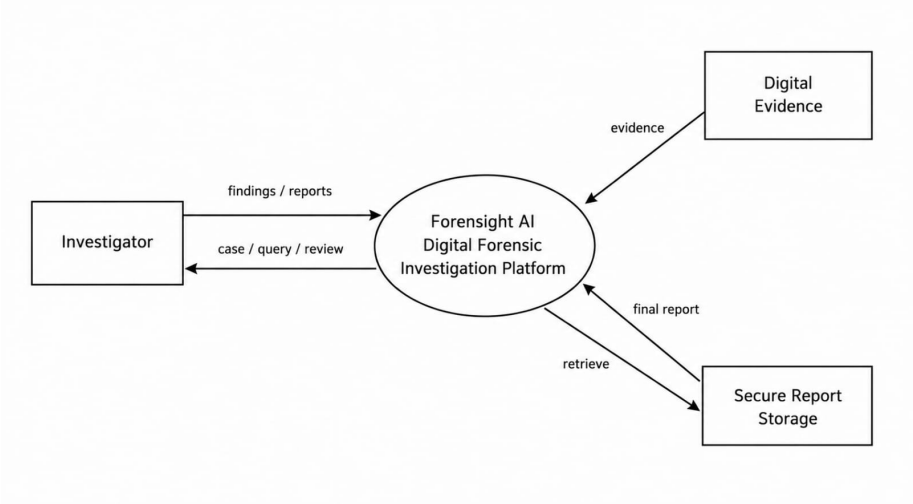
Professional Report Generation
Generation of structured forensic reports

Secure Report Storage
Secure storage and preservation of investigation reports

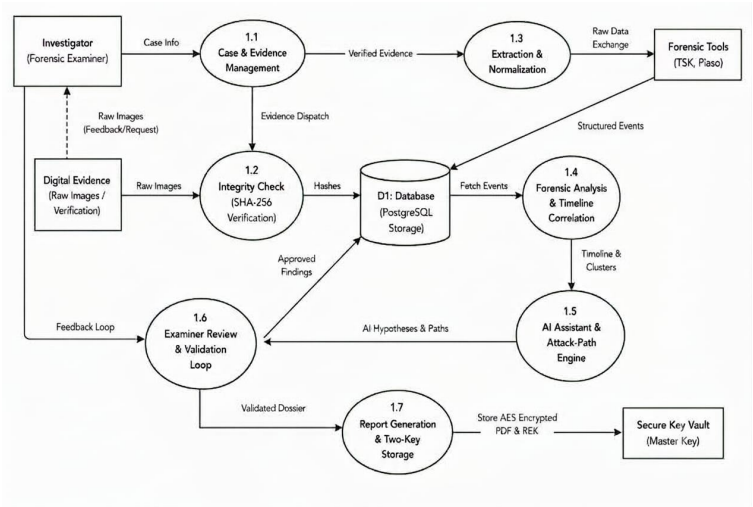
Forensic WorkFlow Diagram



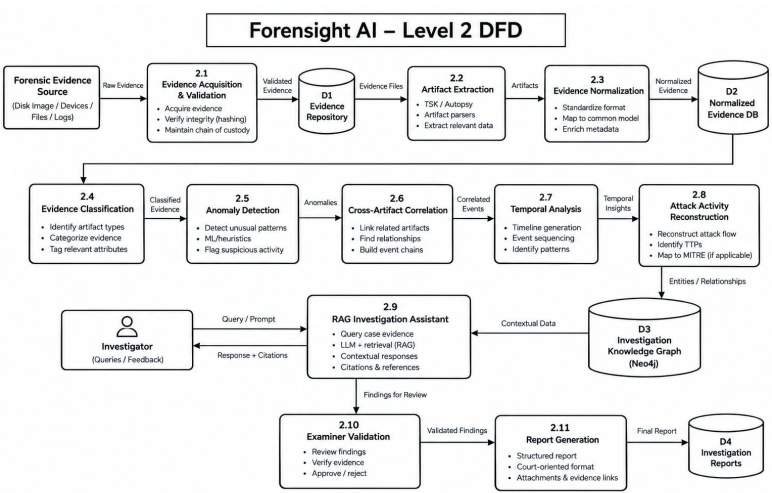
Data Flow Diagram Level-0



Data Flow Diagram Level-1



Data Flow Diagram Level-2



Evidence Verification and Integrity

Evidence Integrity

- Generate SHA-256 hash
- Store original evidence hash
- Verify hash before analysis
- Maintain evidence metadata
- Prevent modification of source evidence

Verification

- Validate extracted artifacts
- Preserve source information
- Verify timestamps
- Track evidence provenance
- Record investigator actions

Forensic Copy $\rightarrow$ **Verification** $\rightarrow$ **Analysis**

AI Investigation Assistant

Role of AI

The AI component operates on structured forensic evidence produced by the investigation engine rather than directly replacing forensic acquisition or extraction.

Investigator Queries

- What happened during this incident?
- Show suspicious PowerShell activity.
- Which evidence indicates persistence?
- Which entities are connected to this IP?

AI-Assisted Tasks

- Summarize incident timeline
- Correlate relevant evidence
- Explain suspicious activity
- Generate investigation report

Structured Evidence → AI Reasoning → Explainable Support

Database Schema



Database Conceptual ER Diagram

Table Name	Primary Key (PK)	Foreign Keys (FK)	Attributes & Data Types	Description
USER	**user_id** (UUID)	None	name, email, password_hash, role (ENUM), created_at	Investigator authentication & role-based access.
CASE	**case_id** (UUID)	**user_id**	case_number, case_name, description, status, created_at	Investigation case container and metadata tracking.
EVIDENCE	**evidence_id** (UUID)	**case_id**	source_name, evidence_type, storage_path, hash_sha256, ingested_at	Ingested disk images, logs, memory dumps & SHA-256 hashes.
ARTIFACT	**artifact_id** (UUID)	**evidence_id**	artifact_type, parser_used, raw_data (JSONB), extracted_at	Extracted forensic artifacts (Registry, MFT, Browser, Logs).
EVENT	**event_id** (UUID)	**artifact_id**	timestamp (UTC), event_type, source_entity, user_account, host_ip, details (JSONB)	Normalized event representation across heterogeneous sources.
FINDING	**finding_id** (UUID)	**case_id**, **event_id**	finding_title, finding_type, description, confidence_score, validation_status, notes	AI-suggested & examiner-validated investigative findings.
ATTACK_NODE	**node_id** (UUID)	**case_id**, **finding_id**	mitre_technique, stage_name, stage_order (INT), description	Reconstructed attack path stages linked to supporting evidence.
REPORT	**report_id** (UUID)	**case_id**	report_name, integrity_hash, encrypted_file_path, encrypted_rek (BYTEA), finalized_at	Finalized forensic dossier, SHA-256 hash & AES-256 REK key.

Hardware Requirements

Component	Minimum	Recommended
CPU	Intel Core i5 / Ryzen 5	Intel Core i7 / Ryzen 7
RAM	8 GB	16 GB or more
Storage	256 GB SSD	512 GB SSD or more
GPU	Integrated	NVIDIA GPU optional
Display	1366×768	Full HD
Network	Optional	Required for external APIs

Development Environment

Windows 11 Host | Kali Linux VM | VirtualBox

Software Requirements

Development

- Python
- FastAPI
- React
- VS Code
- Git / GitHub

Forensics

- The Sleuth Kit
- Autopsy
- Plaso
- YARA
- Volatility

Analysis

- Neo4j
- Cytoscape.js
- MITRE ATT&CK
- Threat Intelligence APIs
- LLM / Local AI

Feasibility Analysis

Type	Status	Justification
Technical	Feasible	Open-source forensic tools, Python frameworks and graph databases are available.
Economic	Highly Feasible	Major components are open-source and can be developed using existing hardware.
Operational	Feasible	Unified visualization can reduce manual effort and support investigators.
Schedule	Feasible	The system can be developed incrementally using modular components.
Legal	Feasible	Analysis can be performed on forensic copies with integrity verification.

Workload Distribution

Member	Primary Responsibility	Major Tasks
Ayush	Forensics and Evidence	Forensic workflow, artifact list, evidence-handling methodology
Rohan	AI/ML & NLP	AI architecture, model pipeline, training dataset requirements
Jerin	Backend, Database & Integration	System architecture, DFD, ER/database design, API structure
Abel	Frontend, Reporting & Documentation	UI wireframes, report template, project documentation

Shared Responsibilities

Integration, testing, documentation, literature review, report preparation and final presentation.

Expected System Output

Investigation Dashboard

- Case information
- Evidence status
- Reconstructed timeline
- Suspicious events
- Entity relationships
- Attack path

Investigation Report

- Incident summary
- Timeline
- Attack path
- Evidence references
- IOCs
- Findings
- Recommendations

Conclusion

- The proposed platform aims to simplify and strengthen digital forensic investigations.
- It combines evidence extraction, timeline reconstruction and graph-based correlation within a unified workflow.
- Cryptographic verification helps preserve evidence integrity and traceability.
- Attack-path reconstruction provides an intuitive representation of incident progression.
- AI-assisted analysis can help investigators summarize evidence, answer investigation queries and generate reports.
- The modular architecture allows future integration of memory forensics, threat intelligence and advanced cybersecurity analytics.

From Raw Evidence to an Explainable Investigation Narrative

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THANK YOU

Questions & Discussion